

- 1 (a)** Find the set of values of k for which the system of equations

$$x + 2y + 3z = 1,$$

$$kx + 4y + 6z = 0,$$

$$7x + 8y + 9z = 3,$$

has a unique solution.

[3]

[illegible]

- (b) Interpret the situation geometrically in the case where the system of equations does not have a unique solution. [2]

[illegible]